Supplementary Table S3

Complete PRISMA Study-Selection and Eligibility Audit

Graph Neural Networks for $PM_{2.5}$ Forecasting: Review on Recent Advancements, Challenges, and Outlooks

Scope of this supplement. The table reports the recorded identification, deduplication, title screening, abstract screening, full-text retrieval, eligibility decisions and final inclusion for the 69-study review. Counts are from the author's original screening pathway, not the later exploratory re-screening. Record-level ledgers (2,180 initial records, 1,937 title decisions, 102 abstract decisions, 93 full-text candidates and 69 included studies) accompany the editable spreadsheet.

Panel A. Complete PRISMA selection sequence

Stage	Recorded decision / flow category	n	Primary evidence
Identification	Scopus after database filtering	391	Database export; S1
Identification	ScienceDirect Search 1 after database filtering	1092	Search string sheet; S1
Identification	ScienceDirect Search 2 after database filtering	81	Search string sheet; S1
Identification	SpringerLink after database filtering	616	Database export; S1
Identification	**All database-export records before deduplication**	**2180**	391 + 1,092 + 81 + 616
Deduplication	Additional duplicate records removed	243	DOI-first/title fallback grouping; 2,180 - 1,937
Deduplication	**Unique records remaining**	**1937**	1937 deduplicated records
Title screening	Unique records screened by title	1937	Historical title screening register
Title screening	Excluded at title screening	1835	T1-T8 reason categories
Title screening	Included/clearly eligible to advance	52	Historical title decision: Include
Title screening	Uncertain/advanced conservatively	50	Historical title decision: Uncertain
Abstract screening	**Abstracts screened**	**102**	52 Include + 50 Uncertain titles
Abstract screening	Excluded at abstract screening	9	2 T2; 7 T6
Abstract screening	Included/clearly eligible to advance	72	Corrected abstract register
Abstract screening	Uncertain/advanced conservatively	21	Corrected abstract register
Full-text retrieval	**Full-text reports sought**	**93**	72 abstract Include + 21 Uncertain
Full-text retrieval	Reports not retrieved	21	Author-confirmed; not assessed for eligibility
Full-text eligibility	**Full-text reports retrieved and assessed**	**72**	93 - 21
Full-text eligibility	Excluded: no PM2.5 forecasting outcome (NO2 target)	2	EX4; S2
Full-text eligibility	Excluded: outside 2019-2026 assigned journal-year rule	1	EX2; S2; AirGFFormer 2027
Full-text eligibility	Total full-text reports excluded	3	2 EX4 + 1 EX2
Inclusion	**Included studies in synthesis**	**69**	72 - 3

Source-filter note. The search-string sheet records 2,821 entries before database-specific filters (449 Scopus, 1,599 and 108 ScienceDirect, 665 SpringerLink). Their post-filter combined output is 2,180 records. The 641 filter-stage removals are not the 243 duplicate removals, which occurred after merging the exported database records. Search expressions are detailed in Supplementary Table S1.

Panel B. Title-screening exclusion reasons ($n = 1{,}835$)

Code	Primary reason recorded	Excluded (n)	Share of exclusions
T1	Not air pollution/air quality	943	51.4%
T2	No qualifying GNN or graph-learning approach identifiable	172	9.4%
T3	Wrong application domain	285	15.5%
T4	Indoor-only, occupational, in-vehicle, underground or similar setting	11	0.6%
T5	Review or other secondary study	117	6.4%
T6	Wrong primary forecast pollutant or target (no eligible PM2.5 outcome)	56	3.1%
T7	Conference paper	232	12.6%
T8	No future forecasting task identified	19	1.0%
Total	**Title-stage exclusions**	**1,835**	**100.0%**

Retained for abstract screening: 52 Include + 50 Uncertain = 102. The T1–T8 labels are the primary reasons from the historical title-screening register; only one primary reason is assigned to each excluded record.

Panel C. Abstract screening and full-text eligibility accounting

Screening component	Recorded category or reason	n
Title-screened records advanced	Abstracts assessed	102
Abstract screening	Include (eligible to seek full text)	72
Abstract screening	Uncertain (advanced for full-text clarification)	21
Abstract screening	Exclude: T2, graph-learning method not established	2
Abstract screening	Exclude: T6, $PM_{2.5}$-specific prediction target not established	7
Full-text retrieval	Reports sought; includes both abstract Include and Uncertain	93
Full-text retrieval	Reports not retrieved (author-confirmed; eligibility unassessed)	21
Full-text eligibility	Reports retrieved and assessed	72
Full-text eligibility	EX4: NO_2-focused studies without eligible $PM_{2.5}$ forecast	2
Full-text eligibility	EX2: assigned 2027 publication year outside applied window	1
Final synthesis	Included studies	**69**

Panel D. Record-level disposition of all 93 full-text candidates

ID	Title	DOI	Database membership	Retrieval and eligibility disposition
2	FTE-GCNGRU: PM2.5 prediction integrating station importance and spatiotemporal deep learning model under multi-layer air quality network	10.1016/j.buildenv.2025.113999	ScienceDirect; Scopus	Included in synthesis
8	Long-term prediction method for PM2.5 concentration using edge channel graph attention network and gating closed-form continuous-time neural networks	10.1016/j.psep.2024.06.090	ScienceDirect; Scopus	Included in synthesis
14	Learning spatiotemporal dependencies using adaptive hierarchical graph convolutional neural network for air quality prediction	10.1016/j.jclepro.2024.142541	ScienceDirect; Scopus	Included in synthesis
21	A reinforcement learning-based spatiotemporal dynamic multi-graph ensemble framework for multi-station air quality prediction	10.1016/j.engappai.2026.115234	ScienceDirect; Scopus	Included in synthesis
24	Air Pollution Prediction Via Graph Attention Network and Gated Recurrent Unit	10.32604/cmc.2022.028411	ScienceDirect; Scopus	Included in synthesis
27	Heterogeneous data fusion considering spatial correlations using graph convolutional networks and its application in air quality prediction	10.1016/j.jksuci.2022.04.003	ScienceDirect; Scopus	Included in synthesis
36	Effective PM2.5 concentration forecasting based on multiple spatial–temporal GNN for areas without monitoring stations	10.1016/j.eswa.2023.121074	ScienceDirect; Scopus	Included in synthesis
37	Regional PM2.5 prediction with hybrid directed graph neural networks and Spatio-temporal fusion of meteorological factors	10.1016/j.envpol.2024.125404	ScienceDirect; Scopus	Included in synthesis
42	Continuous spatio-temporal hybrid graph convolutional network for air pollutant prediction	10.1016/j.ins.2026.123344	ScienceDirect; Scopus	Included in synthesis
49	Spatiotemporal graph neural networks for predicting mid-to-long-term PM2.5 concentrations	10.1016/j.jclepro.2023.138880	ScienceDirect; Scopus	Included in synthesis
52	Adaptive Graph Neural Network–transformer model for high-resolution PM2.5 forecasting and spatial extrapolation	10.1016/j.envsoft.2025.106771	ScienceDirect; Scopus	Included in synthesis
54	Spatiotemporal PM2.5 forecasting via dynamic geographical Graph Neural Network	10.1016/j.envsoft.2025.106351	ScienceDirect; Scopus	Included in synthesis
55	Multi-view multi-task spatiotemporal graph convolutional network for air quality prediction	10.1016/j.scitotenv.2023.164699	ScienceDirect; Scopus	Included in synthesis
60	A graph neural network and Transformer-based model for PM2.5 prediction through spatiotemporal correlation	10.1016/j.envsoft.2025.106501	ScienceDirect; Scopus	Included in synthesis
66	AirGFFormer: Air pollution concentration forecasting with graph convolutional networks and frequency transformer	10.1016/j.eswa.2026.134439	ScienceDirect; Scopus	Excluded after full text: 2027 year
69	Adaptive graph-generating jump network for air quality prediction based on improved graph convolutional network	10.1016/j.apr.2025.102488	ScienceDirect; Scopus	Included in synthesis
71	A city-based PM2.5 forecasting framework using Spatially Attentive Cluster-based Graph Neural Network model	10.1016/j.jclepro.2023.137036	ScienceDirect; Scopus	Included in synthesis
72	Exploring urban environmental semantics for air quality prediction using explainable multi-view spatiotemporal graph neural networks	10.1016/j.apgeog.2025.103605	ScienceDirect; Scopus	Excluded after full text: NO2 target
75	Predicting air quality using a multi-scale spatiotemporal graph attention network	10.1016/j.ins.2024.121072	ScienceDirect; Scopus	Included in synthesis
76	A dual-path dynamic directed graph convolutional network for air quality prediction	10.1016/j.scitotenv.2022.154298	ScienceDirect; Scopus	Included in synthesis

Panel D continued (93 full-text candidates)

ID	Title	DOI	Database membership	Retrieval and eligibility disposition
83	Fine-grained PM2.5 prediction in Lanzhou based on the spatiotemporal graph convolutional network	10.1016/j.apr.2023.101993	ScienceDirect; Scopus	Included in synthesis
84	Uncertainty graph convolution recurrent neural network for air quality forecasting	10.1016/j.aei.2024.102651	ScienceDirect; Scopus	Included in synthesis
86	A hybrid spatiotemporal model combining graph attention network and gated recurrent unit for regional composite air pollution prediction and collaborative control	10.1016/j.scs.2024.105925	ScienceDirect; Scopus	Included in synthesis
88	Forecasting PM2.5 concentrations with spatial-temporal hypergraph attention network: Modeling geospatial correlations and latent associations among stations	10.1016/j.eswa.2025.128298	ScienceDirect; Scopus	Included in synthesis
93	Adaptive scalable spatio-temporal graph convolutional network for PM2.5 prediction	10.1016/j.engappai.2023.107080	ScienceDirect; Scopus	Included in synthesis
97	Dynamic graph neural network with adaptive edge attributes for air quality prediction: A case study in China	10.1016/j.heliyon.2023.e17746	ScienceDirect; Scopus	Included in synthesis
102	MGAtt-LSTM: A multi-scale spatial correlation prediction model of PM2.5 concentration based on multi-graph attention	10.1016/j.envsoft.2024.106095	ScienceDirect	Included in synthesis
106	Multi-granularity PM2.5 concentration long sequence prediction model combined with spatial–temporal graph	10.1016/j.envsoft.2025.106400	ScienceDirect; Scopus	Included in synthesis
112	An adaptive hypergraph-based convolution network with dual spatiotemporal attention for PM2.5 forecasting	10.1016/j.eswa.2025.129497	ScienceDirect	Included in synthesis
121	An Adaptive Spatiotemporal Graph Transformer for Multi-Site PM2.5 Multi-Step Forecasting with Non-stationary and Sparse-Aware Method	10.1016/j.jenvman.2026.130515	ScienceDirect; Scopus	Included in synthesis
127	Spatio-temporal fusion of meteorological factors for multi-site PM2.5 prediction: A deep learning and time-variant graph approach	10.1016/j.envres.2023.117286	ScienceDirect; Scopus	Included in synthesis
135	A long-term prediction method for PM2.5 concentration based on spatiotemporal graph attention recurrent neural network and grey wolf optimization algorithm	10.1016/j.jece.2023.111716	ScienceDirect; Scopus	Included in synthesis
171	PM2.5 forecasting under distribution shift: A graph learning approach	10.1016/j.aiopen.2023.11.001	ScienceDirect; Scopus	Included in synthesis
173	72-hour real-time forecasting of ambient PM2.5 by hybrid graph deep neural network with aggregated neighborhood spatiotemporal information	10.1016/j.envint.2023.107971	ScienceDirect	Included in synthesis
176	A graph-based LSTM model for PM2.5 forecasting	10.1016/j.apr.2021.101150	ScienceDirect; Scopus	Included in synthesis
186	Short-term air pollution prediction using graph convolutional neural networks	10.1016/j.techfore.2024.123684	ScienceDirect	Included in synthesis
192	Dynamic graph convolution neural network based on spatial-temporal correlation for air quality prediction	10.1016/j.ecoinf.2022.101736	ScienceDirect; Scopus	Included in synthesis
205	A theory-guided graph networks based PM2.5 forecasting method	10.1016/j.envpol.2021.118569	ScienceDirect; Scopus	Included in synthesis
208	Spatiotemporal integration of GCN and E-LSTM networks for PM2.5 forecasting	10.1016/j.mlwa.2023.100521	ScienceDirect	Included in synthesis
226	Forecasting PM2.5 using hybrid graph convolution-based model considering dynamic wind-field to offer the benefit of spatial interpretability	10.1016/j.envpol.2021.116473	ScienceDirect	Included in synthesis
250	A hybrid model for spatiotemporal forecasting of PM 2.5 based on graph convolutional neural network and long short-term memory	10.1016/j.scitotenv.2019.01.333	ScienceDirect; Scopus	Included in synthesis

Panel D continued (93 full-text candidates)

ID	Title	DOI	Database membership	Retrieval and eligibility disposition
257	A new multi-data-driven spatiotemporal PM2.5 forecasting model based on an ensemble graph reinforcement learning convolutional network	10.1016/j.apr.2021.101197	ScienceDirect; Scopus	Included in synthesis
275	Distance adaptive graph convolutional gated network-based smart air quality monitoring and health risk prediction in sensor-devoid urban areas	10.1016/j.scs.2023.104445	ScienceDirect	Included in synthesis
282	Enhanced PM2.5 prediction with a dynamic wind-driven graph fusion model incorporating inter-station pollutant transport	10.1016/j.atmosenv.2024.121012	ScienceDirect	Included in synthesis
297	A new ensemble spatio-temporal PM2.5 prediction method based on graph attention recursive networks and reinforcement learning	10.1016/j.chaos.2022.112405	ScienceDirect; Scopus	Included in synthesis
423	A PM2.5 spatiotemporal prediction model based on mixed graph convolutional GRU and self-attention network	10.1016/j.envpol.2025.125748	ScienceDirect	Included in synthesis
1099	PM2.5 prediction based on dynamic spatiotemporal graph neural network	10.1007/s10489-024-05801-7	Scopus; Springer	Not retrieved; unassessed
1102	An Ensemble Model with Adaptive Variational Mode Decomposition and Multivariate Temporal Graph Neural Network for PM2.5 Concentration Forecasting	10.3390/su142013191	Scopus	Included in synthesis
1103	A Spatio-Temporal Graph Convolutional Network for Air Quality Prediction	10.3390/su15097624	Scopus	Included in synthesis
1109	Air quality forecasting using a spatiotemporal hybrid deep learning model based on VMD–GAT–BiLSTM	10.1038/s41598-024-68874-x	Scopus	Included in synthesis
1113	For the aged: A novel PM2.5 concentration forecasting method based on spatial-temporal graph ordinary differential equation networks in home-based care parks	10.3389/fenvs.2022.956020	Scopus	Included in synthesis
1128	Multi-scale dynamic graph neural network for PM2.5 concentration prediction in regional station cluster	10.1371/journal.pone.0338392	Scopus	Included in synthesis
1140	A Novel Model for Predicting PM2.5 Concentrations Utilizing Graph Convolutional Networks and Transformer	10.1109/access.2025.3541774	Scopus	Included in synthesis
1149	AirGraphNet: a novel spatio-temporal graph attention model for PM2.5 forecasting	10.1007/s00500-025-10843-1	Scopus; Springer	Not retrieved; unassessed
1160	AGATNet: An Adaptive Graph Attention Network for Bias Correction of CMAQ-Forecasted PM2.5 Concentrations Over South Korea	10.1029/2024jh000244	Scopus	Included in synthesis
1161	A Spatiotemporal graph attention network for PM2.5 forecasting using multi-source data	10.1007/s11869-025-01818-0	Scopus; Springer	Not retrieved; unassessed
1168	A Graph Attention Recurrent Neural Network Model for PM2.5 Prediction: A Case Study in China from 2015 to 2022	10.3390/atmos15070799	Scopus	Included in synthesis
1186	Predicting short-term PM2.5 concentrations at fine temporal resolutions using a multi-branch temporal graph convolutional neural network	10.1080/13658816.2024.2310737	Scopus	Not retrieved; unassessed
1199	A Dynamic Global–Local Spatiotemporal Graph Framework for Multi-City PM2.5 Long-Term Forecasting	10.3390/rs17162750	Scopus	Included in synthesis
1200	A Lightweight Spatiotemporal Graph Framework Leveraging Clustered Monitoring Networks and Copula-Based Pollutant Dependency for PM2.5 Forecasting	10.3390/land14081589	Scopus	Included in synthesis
1208	A short-term prediction approach for air quality based on Triple cross-attention and self-loop normalized Graph Attention Network	10.1007/s12145-025-01883-0	Scopus; Springer	Not retrieved; unassessed

Panel D continued (93 full-text candidates)

ID	Title	DOI	Database membership	Retrieval and eligibility disposition
1213	A Novel Spatio-Temporal Graph Convolutional Network with Attention Mechanism for PM2.5 Concentration Prediction	10.3390/make7030088	Scopus	Included in synthesis
1215	Enhancing Air Quality Forecasting: A Novel Spatio-Temporal Model Integrating Graph Convolution and Multi-Head Attention Mechanism	10.3390/atmos15040418	Scopus	Included in synthesis
1219	Temporal Difference-Based Graph Transformer Networks For Air Quality PM2.5 Prediction: A Case Study in China	10.3389/fenvs.2022.924986	Scopus	Included in synthesis
1223	Dual-channel spatial–temporal difference graph neural network for PM 2.5 forecasting	10.1007/s00521-022-08036-0	Scopus; Springer	Not retrieved; unassessed
1231	GAT-EGRU: A Deep Learning Prediction Model for PM2.5 Coupled with Empirical Modal Decomposition Algorithm	10.1007/s11518-023-5562-z	Scopus; Springer	Not retrieved; unassessed
1232	A novel spatiotemporal multigraph convolutional network for air pollution prediction	10.1007/s10489-022-04418-y	Scopus; Springer	Not retrieved; unassessed
1237	Forecasting PM2.5 Concentration in India Using a Cluster Based Hybrid Graph Neural Network Approach	10.1007/s13143-022-00291-4	Scopus; Springer	Not retrieved; unassessed
1248	Multi-Source PM2.5 Prediction Model Based on Fusion of Graph Attention Networks and Multiple Time Periods	10.1109/access.2024.3390934	Scopus	Included in synthesis
1257	Human migration-based graph convolutional network for PM2.5 forecasting in post-COVID-19 pandemic age	10.1007/s00521-022-07876-0	Scopus; Springer	Not retrieved; unassessed
1264	A Novel Framework for Air Quality Forecasting Using Graph Convolutional Network-Based Time Series Decomposition	10.15244/pjoes/187607	Scopus	Included in synthesis
1266	ST-BernT: A Spatiotemporal λ-Bernstein Graph Convolutional Network With Transformer for Multisite Air Quality Prediction in Distributed Unmanned Agent Systems	10.1109/jiot.2025.3617922	Scopus	Not retrieved; unassessed
1278	Boosting PM2.5 short-term prediction by coupling graph attention and koopman networks	10.1007/s43762-026-00250-z	Scopus; Springer	Included in synthesis
1286	PM2.5 Concentration Forecasting in Industrial Parks Based on Attention Mechanism Spatiotemporal Graph Convolutional Networks	10.1155/2021/7000986	Scopus	Included in synthesis
1289	PM2.5 prediction and its influencing factors in the Beijing-Tianjin-Hebei urban agglomeration using spatial temporal graph convolutional networks	10.1088/2515-7620/ade1aa	Scopus	Included in synthesis
1290	Benchmarking Deep Learning for PM2.5 Forecasting in IoT-Based Smart Cities: GCN Spatial Encoding and Transformer Temporal Modeling	10.14569/ijacsa.2026.0170595	Scopus	Included in synthesis
1300	On hourly prediction of PM2.5 using spatial–temporal graph convolutional network	10.1007/s12145-025-01890-1	Scopus; Springer	Not retrieved; unassessed
1304	Graph Neural Network for Air Quality Prediction: A Case Study in Madrid	10.1109/access.2023.3234214	Scopus	Excluded after full text: NO2 target
1312	Nationwide Air Pollution Forecasting with Heterogeneous Graph Neural Networks	10.1145/3637492	Scopus	Not retrieved; unassessed
1320	MTLPM: a long-term fine-grained PM2.5 prediction method based on spatio-temporal graph neural network	10.1007/s10661-024-13407-2	Scopus; Springer	Not retrieved; unassessed
1321	Deep Spatio-Temporal Graph Network with Self-Optimization for Air Quality Prediction	10.3390/e25020247	Scopus	Included in synthesis
1325	Semi-Supervised Air Quality Forecasting via Self-Supervised Hierarchical Graph Neural Network	10.1109/tkde.2022.3149815	Scopus	Not retrieved; unassessed

Panel D continued (93 full-text candidates)

ID	Title	DOI	Database membership	Retrieval and eligibility disposition
1329	Simulating the air quality impact of prescribed fires using graph neural network-based PM 2.5 forecasts	10.1017/eds.2025.4	Scopus	Included in synthesis
1342	MGC-LSTM: a deep learning model based on graph convolution of multiple graphs for PM2.5 prediction	10.1007/s13762-022-04553-6	Scopus; Springer	Not retrieved; unassessed
1354	Spatio-temporal graph neural network for inter-city air quality forecasting	10.1007/s13762-025-06850-2	Scopus; Springer	Not retrieved; unassessed
1358	An adaptive adjacency matrix-based graph convolutional recurrent network for air quality prediction	10.1038/s41598-024-55060-2	Scopus; Springer	Included in synthesis
1359	Dynamic mode decomposition and short-time prediction of PM2.5 using the graph Neural Koopman network	10.1080/13658816.2024.2408749	Scopus	Not retrieved; unassessed
1374	CMDNet: A Cross-Modality Spatiotemporal Graph Network for Enhanced Air Pollution Prediction With High-Resolution Satellite Data	10.1109/tgrs.2025.3569594	Scopus	Not retrieved; unassessed
1413	Interpretable spatio-temporal graph neural networks for air quality prediction under real-world multi-station monitoring data	10.1038/s41598-026-66494-1	Springer	Included in synthesis
1449	Neural CDE-GAT framework for spatio-temporal air pollution prediction	10.1038/s41598-026-60629-0	Springer	Included in synthesis
1664	Near-surface PM2.5 prediction combining the complex network characterization and graph convolution neural network	10.1007/s00521-021-06300-3	Springer	Not retrieved; unassessed
1731	Multi-scale spatiotemporal graph convolution network for air quality prediction	10.1007/s10489-020-02054-y	Springer	Not retrieved; unassessed
109	A graph-integrated latent ensemble random vector functional link without direct links framework for drift-resilient online multi-step fine particulate matter forecasting	10.1016/j.asoc.2026.115470	ScienceDirect	Included in synthesis

Record-level totals: 69 included; 21 not retrieved; 3 excluded after full-text assessment. The spreadsheet retains the complete original titles, journal names, authors, eligibility reasons and abstract-stage provenance.

Panel E. Three reports excluded following full-text assessment

ID	Study	DOI	Code	Primary eligibility reason
S2-01	Graph Neural Network for Air Quality Prediction: A Case Study in Madrid	10.1109/ACCESS.2023.3234214	EX4	The full-text assessment identified NO2 concentration as the forecasting target; PM2.5 concentration was not forecast.
S2-02	Exploring urban environmental semantics for air quality prediction using explainable multi-view spatiotemporal graph neural networks	10.1016/j.apgeog.2025.103605	EX4	The London analysis evaluated NO2 prediction; a separate future PM2.5 forecasting outcome was not established.
S2-03	AirGFFormer: Air pollution concentration forecasting with graph convolutional networks and frequency transformer	10.1016/j.eswa.2026.134439	EX2	The article is assigned journal publication year 2027, outside the applied 2019-2026 publication-year criterion; PM2.5 forecasting was otherwise reported.

Methods and evidentiary boundaries

- The original search exports record 2,180 post-filter database records, from which 243 additional copies were removed. DOI-first matching with a normalized-title fallback reconstructs 1,937 unique groups (1,709 groups of one record, 213 groups of two and 15 groups of three).
- The historical title-screening register supports 1,835 exclusions under T1–T8; the corrected 102-abstract register supports nine exclusions. A related latent-feature graph-learning article was re-adjudicated at the abstract stage to reconcile 93 full-text candidates.
- The 21 reports identified as not retrieved reflect author-confirmed inability to obtain the full texts. No independent dated retrieval-attempt log was supplied; retrieval outcomes should not be interpreted as substantive eligibility rejections.
- Three assessed reports were excluded: two lacked an eligible $PM_{2.5}$ forecast target and one was assigned a journal year of 2027 under the applied eligibility window. No exclusion for low predictive performance was used.
- Screening counts and decisions are reproduced from the supplied original/corrected registers; this supplementary table is an audit presentation, not proof that screening was independently performed by multiple reviewers or that the original primary study PDFs were revalidated one by one.

Accompanying files. Complete ledgers for identification (2,180 entries), title screening (1,937 decisions), abstract screening (102 decisions), full-text retrieval (93 candidates; including 21 not retrieved), eligibility exclusions (3 reports) and final inclusions (69 studies) are available as separately labelled spreadsheet worksheets and CSV files. Database-specific search strings are in Supplementary Table S1; detailed exclusion descriptions are in Supplementary Table S2.